

CivicTrack System

Phase 3: Design Specification

Blueprint Before Coding

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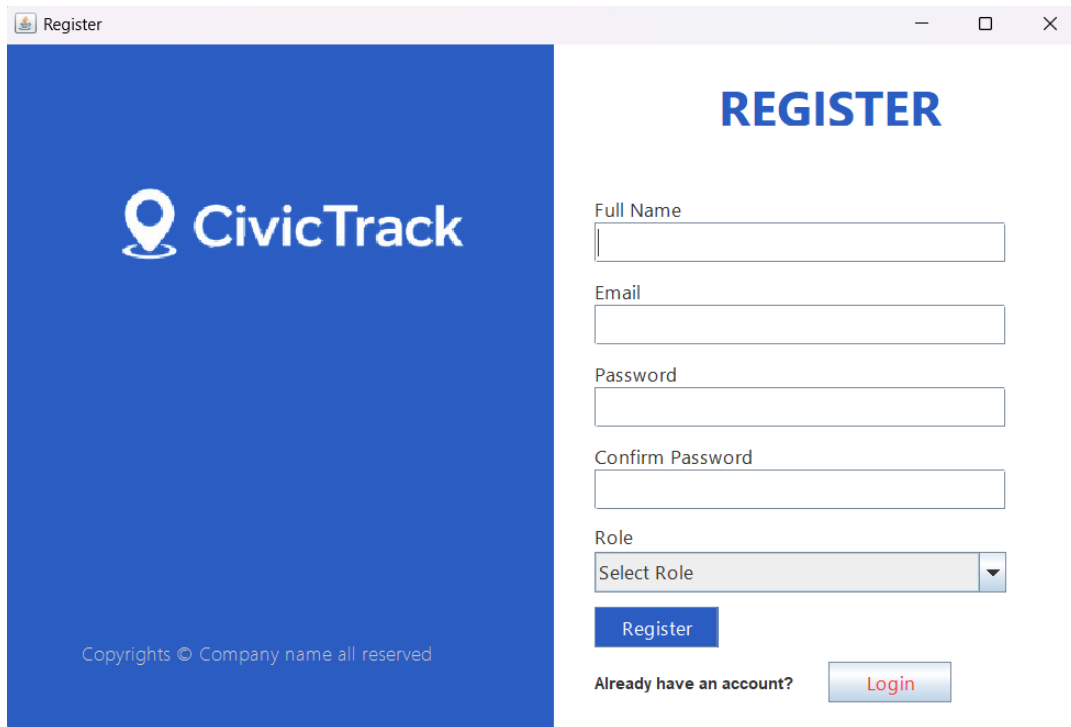
Bonolo Sesinyi 202464086

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1. GUI Design Documentation(Interface Mock ups)

1.1. SCREEN 1: REGISTRATION SCREEN



A browser window titled "Register" with standard window controls. The left side features a blue background with the "CivicTrack" logo (a location pin icon) and the text "Copyrights © Company name all reserved". The right side has a white background with the heading "REGISTER" in blue. Below the heading are five form fields: "Full Name", "Email", "Password", "Confirm Password", and "Role". The "Role" field is a dropdown menu with "Select Role" as the placeholder. Below these fields is a blue "Register" button. At the bottom, there is a link "Already have an account?" followed by a "Login" button.

Register

REGISTER

Full Name

Email

Password

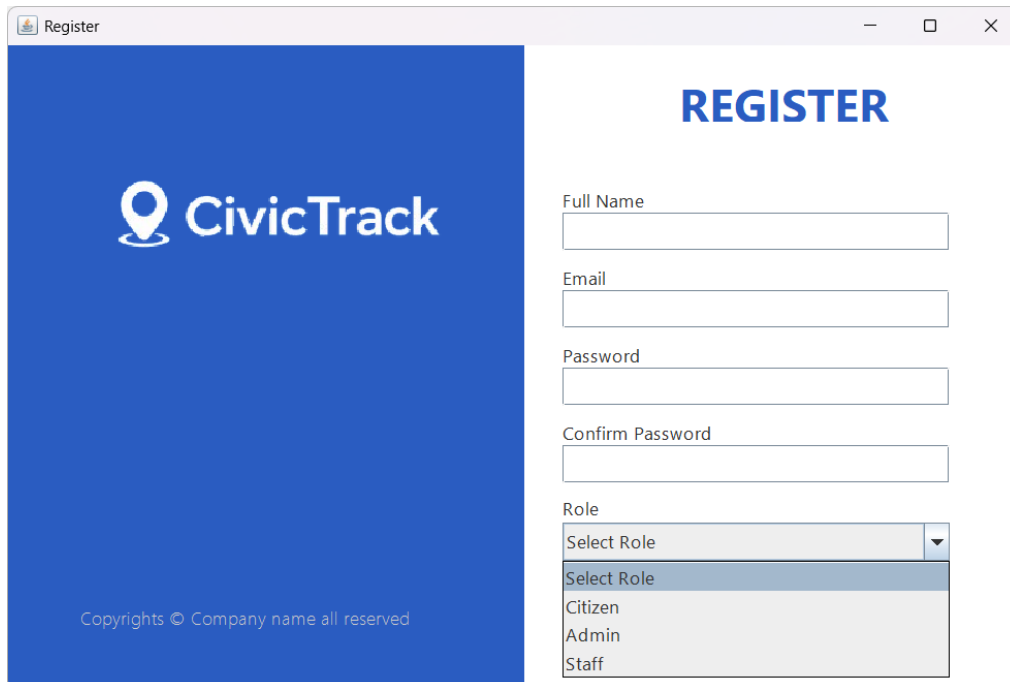
Confirm Password

Role
Select Role ▼

Register

Already have an account? [Login](#)

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A browser window titled "Register" with standard window controls. The left side features a blue background with the "CivicTrack" logo (a location pin icon) and the text "Copyrights © Company name all reserved". The right side has a white background with the heading "REGISTER" in blue. Below the heading are five form fields: "Full Name", "Email", "Password", "Confirm Password", and "Role". The "Role" field is a dropdown menu with "Select Role" as the placeholder, and its menu is open, showing the options "Select Role", "Citizen", "Admin", and "Staff". Below these fields is a blue "Register" button. At the bottom, there is a link "Already have an account?" followed by a "Login" button.

Register

REGISTER

Full Name

Email

Password

Confirm Password

Role
Select Role ▼

Select Role
Citizen
Admin
Staff

Register

Already have an account? [Login](#)

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Description:

The Registration screen allows new users to create an account in the CivicTrack system. This screen is essential for onboarding citizens, municipal staff, and administrators who need access to the system. The Registration screen displays the CivicTrack logo at its center top, with "REGISTER" written clearly, keeping up with the branding of the Login screen.

There are five input fields for the user to fill out:

Element	Type	Purpose
Full Name field	Text input	Captures the user's complete legal name for identification and communication
Email field	Text input	Captures user's email address, which becomes their unique login credential and receives system notifications
Password field	Password input (masked)	Captures a secure password; characters are hidden during typing for security
Confirm Password field	Password input (masked)	Ensures the user typed their intended password correctly; system will compare both fields
Role dropdown	JComboBox (dropdown menu)	Allows users to select their specific access level (Citizen, Staff, or Admin) to determine system permissions.

- The Role drop-down is very important since it will determine what rights and dashboard that the user gets once, he/she logs in to the system:
- Citizen – Submit issues, check progress on issues raised, get notifications
- Admin – Has full access, manages users, controls all issues, performs analysis
- Staff – City staff with rights to look at issues and change status

- Underneath the input boxes, the Register button appears. Clicking on it triggers the validation process:
 1. All fields must be filled
 2. E-mail address must be in valid format (e.g., name@domain.com)
 3. Password and Password Confirm should be identical
 4. Password must be at least six characters long
 5. User must select a role
- If the validation is successful, the system will add a new account entry into the database consisting of the entered email, password (in encrypted format), the user's full name, and role. The user will then be directed to the Login page. Below the Register button, there is a text that says, "Already have an account.." and then the Login button (or link). The footer in the copyright section consists of the same text as found in the Login page: "Copyright © CivicTrack. All rights reserved."

Navigation: Flow between screens

User Action	Destination Screen
Click Register button (successful validation)	Login screen
Click Register button (failed validation)	Stay on Registration screen (error messages shown)
Click Login button (or "Already have an account" link)	Login screen
Click Role dropdown	Dropdown expands showing: Citizen, Admin, Staff (stays on same screen)

1.2. SCREEN 1: LOGIN SCREEN

LOGIN

LOGIN

Email

Password

LogIn

Don't have an account? Register

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Description:

- The Login screen is the gateway into the CivicTrack application. The Login screen consists of the CivicTrack logo at the centre of the screen, and “LOGIN” written at the top of the screen.

Users encounter two input fields:

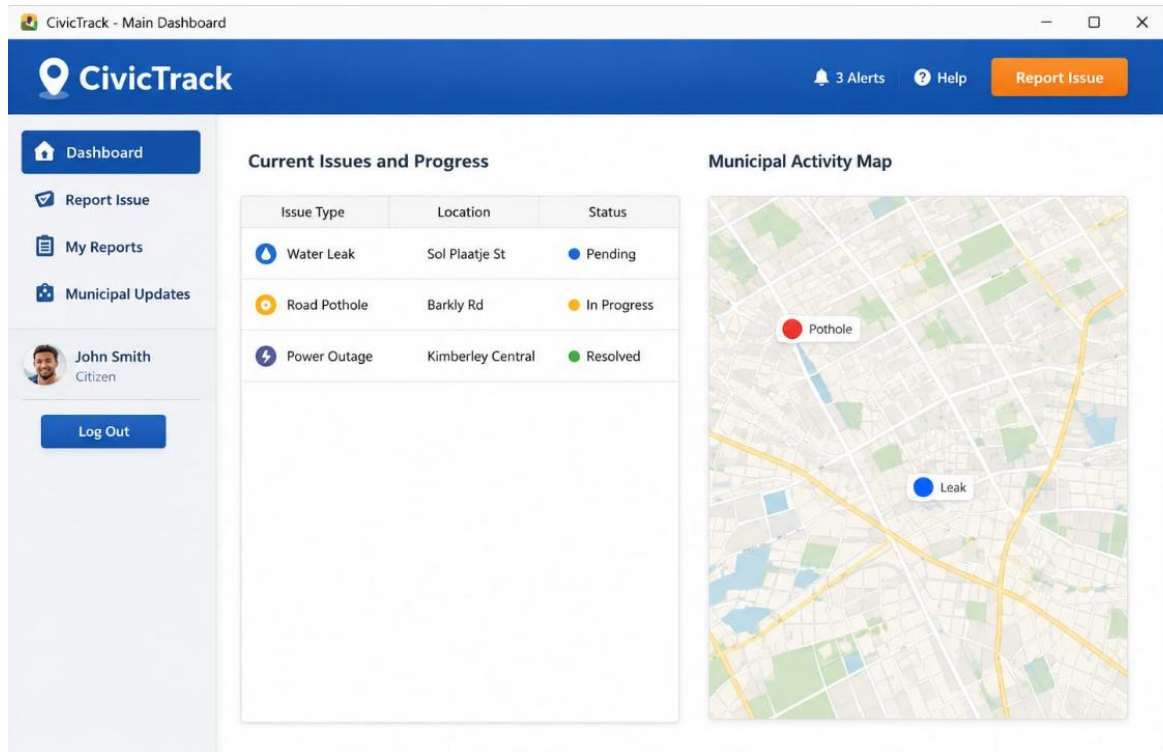
Element	Type	Type
Email field	Text input	Allows user to enter their registered email address, which serves as their unique username
Password field	Password input (masked)	Allows user to enter their password securely; all characters are hidden with dots or asterisks

- After the input boxes, the Login button will be visible in a contrasting color. Clicking this will cause the system to validate the credentials entered against those stored in the database. The system will check not only the email and password but also the user's role (Citizen, Admin, or Staff). Successful validation will lead to redirection to the role-specific dashboard:
 - **Citizen** → Citizen Dashboard (report issues, track progress)
 - **Admin** → Admin Panel (manage users, oversee all reports)
 - **Staff** → Staff Dashboard (assign technicians, update issue status)
- When the credentials are not valid, an error message pops up indicating to retry or reset the password. Under the login button, there is a text saying, "Not registered?" followed by a button for registration. This indicates to new users that they can register themselves through the register button. The copyright section on the bottom consists of "Copyright © CivicTrack. All rights reserved."

Navigation: Flow between screens:

User Action	Destination Screen
Click Login button (valid Citizen credentials)	Citizen Dashboard
Click Login button (valid Admin credentials)	Admin Panel
Click Login button (valid Staff credentials)	Staff Dashboard
Click Login button (invalid credentials)	Stay on Login screen (error message shown)
Click Register button	Registration screen

1.3. SCREEN 3: MAIN DASHBOARD SCREEN



Description:

The Main Dashboard serves as the primary interface for citizens in Kimberley to interact with municipal services. It is designed with a three-section grid layout: a high-contrast top navigation bar, a persistent sidebar for navigation, and a central workspace.

User Interface Components:

- **Report Issue Button:** A prominent orange action button located in the top-right to ensure immediate visibility and quick access for users.
- **Current Issues Table:** Displays the "Issue Type," "Location," and "Status" (e.g., Pending, In Progress, Resolved) to provide real-time feedback to the user.
- **Municipal Activity Map:** A coordinate-based grid visualizing reported issues through color-coded pins (Red for Potholes, Blue for Leaks).

Navigation: Flow between screens

- **Action:** Clicking the "Report Issue" button redirects the user to the Report Issue Screen.
- **Action:** Clicking items in the Sidebar (e.g., My Reports, Updates) allows the user to toggle between different views without losing their session context.

HCI Principles Applied:

- **Visibility of System Status:** The "Current Issues" table ensures users know exactly what is happening with their reports at all times.
- **Aesthetic and Minimalist Design:** The use of white space and a limited color palette (Navy and Orange) reduces cognitive load, helping users find the "Report" function instantly.

1.4. SCREEN 4: REPORT ISSUE SCREEN

The screenshot displays the 'Report a Municipal Issue' screen in the CivicTrack application. The interface is clean and user-friendly, with a blue header bar containing the CivicTrack logo and navigation links for Alerts, Help, and Dashboard. A sidebar on the left provides quick access to the Dashboard, Report Issue (the current screen), My Reports, and Municipal Updates. The main content area is titled 'Report a Municipal Issue' and contains a form for submitting a report. The form includes a dropdown menu for 'Issue Category' (currently set to 'Pothole'), text input fields for 'Description' and 'Location', and an 'Upload Photo' section with a 'Choose File' button and the text 'No file selected'. A user profile for John Smith, a Citizen, is displayed in the sidebar with a 'Log Out' button.

Description:

The Users can submit civic-related issues to the system on the Report Issue screen. To ensure clarity of use, it is designed as a user-friendly form interface with clearly defined input fields and actions.

User Interface Elements

Sidebar Navigation Panel

- Dashboard-Gets to the basic overview interface.
- Report Issue-This opens the active screen of the current form.
- My Reports- Identifies reports that had already been submitted.
- Updates- Provides warnings and municipal status updates.

Main Form Area:

- Type of Issue (Dropdown Menu)- enables users to assign problem's category (e.g., Water leak, potholes, illegal dumping).
- Field for Description (Text Input)- enables users to provide particular about the issues
- Field for Location (Text Input)- Used to correctly recognize individuals by providing their street address or a nearby landmark.
- Button for submitting photographs- allow users to add image as evidence
- Press the submit button- forwards the report to the system for handling.

Navigation: Flow between screens

- Dashboard → Report Issue
Users access this screen by selecting “Report Issue” from the sidebar.
- Report Issue → My Reports
After submitting, users can view their reports in the “My Reports” section.
- Report Issue → Updates
Users can track the progress or status of reported issues through the “Updates” section

1.5. SCREEN 5: TRACK ISSUE SCREEN

The screenshot displays the 'Track Municipal Issue' screen within the CivicTrack application. The interface is clean and functional, with a clear navigation path. The sidebar provides quick access to other key features, while the main area is dedicated to tracking a specific issue by entering a reference number.

Description:

The Track Issue Screen allows citizens to monitor the progress of municipal issues they previously reported. The interface is designed using a clean three-section layout consisting of a top navigation bar, a left-side navigation panel, and a central tracking workspace. The page uses a blue-and-white colour scheme to create a professional and easy-to-use environment while helping users quickly locate tracking features.

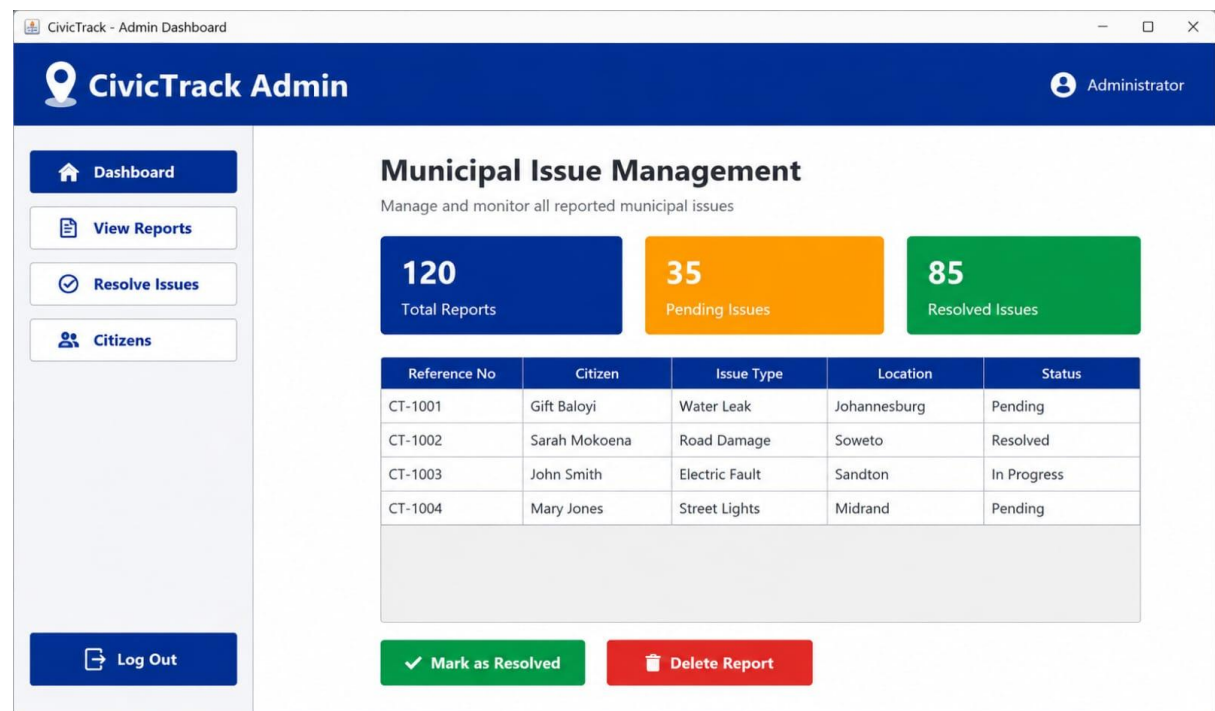
- **User Interface Components:**

- **Top Navigation Bar:** Displays the CivicTrack logo and a Help option, providing branding and quick assistance access for users.
 - **Sidebar Navigation Panel:** Located on the left side of the screen and contains navigation buttons such as Dashboard, Track Issue, My Reports, and Log Out for smooth movement between system pages.
 - **Reference Number Input Field:** A large text field positioned in the centre where users enter their issue reference number to search for updates on their reported problem.
 - **Track Issue Button:** A bold blue action button placed below the input field that initiates the tracking process once clicked.
 - **Results Display Area:** A large text area below the button used to display issue details such as issue type, reported location, progress status, and municipal feedback.
 - **Log Out Button:** Positioned at the bottom of the sidebar, allowing users to securely exit the system session.

Navigation: Flow between screens:

- **Action:** Clicking the “Track Issue” button retrieves and displays the status of the municipal issue linked to the entered reference number.
- **Action:** Selecting “Dashboard” redirects the user back to the main dashboard interface.
- **Action:** Selecting “My Reports” opens the page containing all previously submitted reports by the user.
- **Action:** Clicking “Log Out” securely ends the user session and redirects them to the login screen.
- **Action:** Clicking “Help” provides guidance or support information for using the tracking system.

1.5. SCREEN 5: ADMIN CONTROL PANEL SCREEN



Description:

The Admin Dashboard serves as the main management interface for municipal administrators to monitor, manage, and resolve community-reported issues within the CivicTrack system. The interface follows a structured three-section layout consisting of a top navigation bar, a sidebar navigation menu, and a central management workspace. The dashboard uses a professional blue, white, green, and orange colour scheme to improve readability and highlight important system statistics.

User Interface Components:

- **Top Navigation Bar:** Displays the CivicTrack Admin logo and administrator profile information to identify the current logged-in user.
- **Sidebar Navigation Panel:** Located on the left side of the screen and contains navigation buttons such as Dashboard, View Reports, Resolve Issues, Citizens, and Log Out for efficient system navigation.
- **Statistics Summary Cards:** Three coloured information cards display important municipal data including:

- Total Reports
- Pending Issues
- Resolved Issues

These cards provide administrators with quick system insights and real-time monitoring information.

• **Reports Management Table:** A structured table displaying municipal issue information including:

- Reference Number
- Citizen Name
- Issue Type
- Location
- Status

The table allows administrators to monitor all submitted reports in one centralized view.

• **Mark as Resolved Button:** A green action button used to update the status of selected municipal issues once they have been completed.

• **Delete Report Button:** A red action button allowing administrators to remove invalid, duplicate, or unnecessary reports from the system database.

• **Log Out Button:** Positioned at the bottom of the sidebar to securely end the administrator session.

Navigation: Flow between screen:

- **Action:** Clicking “Dashboard” reloads the main administrative overview screen.
- **Action:** Clicking “View Reports” displays all submitted municipal reports in detail.
- **Action:** Clicking “Resolve Issues” allows administrators to manage and update unresolved municipal problems.
- **Action:** Clicking “Citizens” opens citizen-related information and user management features.
- **Action:** Selecting a row in the reports table enables administrative actions such as resolving or deleting reports.
- **Action:** Clicking “Mark as Resolved” changes the selected issue status from Pending or In Progress to Resolved.
- **Action:** Clicking “Delete Report” permanently removes the selected issue report from the system.

1.6. User Feedback and Design Iterations.

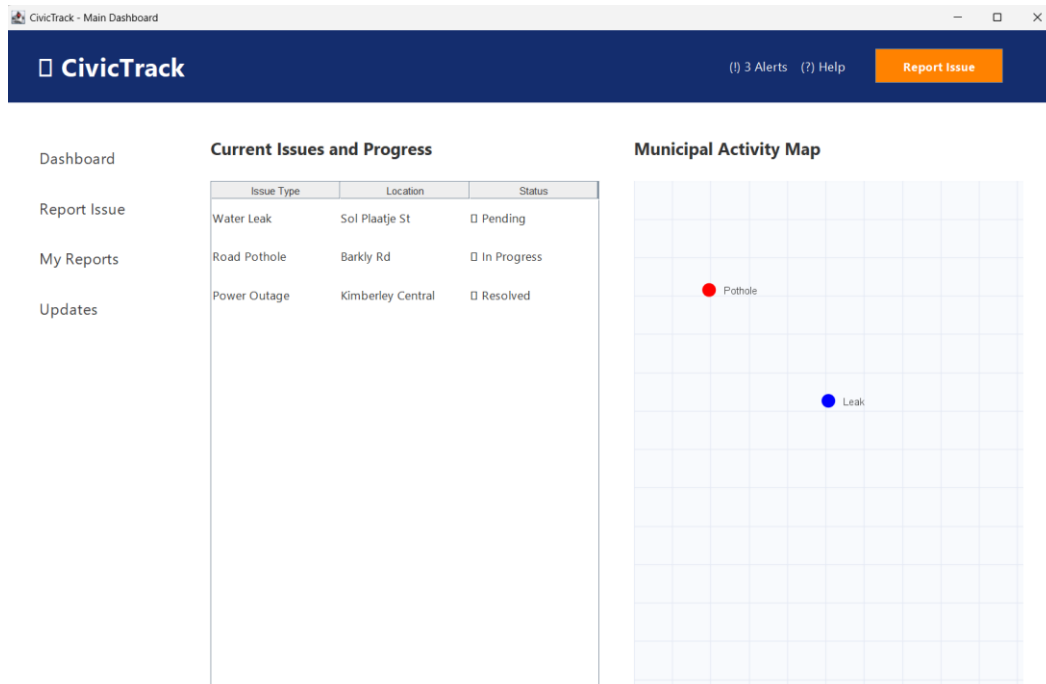
To improve the CivicTrack system, we conducted usability testing with a group of potential users in Kimberley. We observed how they interacted with our initial mock-ups and recorded their feedback to make the system more user-friendly.

Summary of User Testing & Improvements

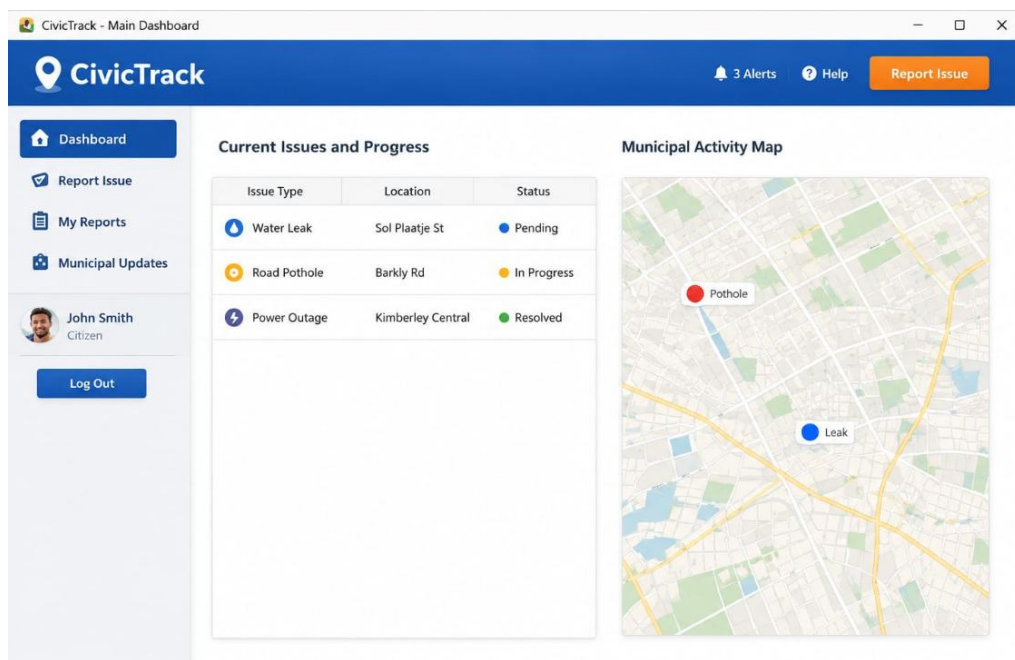
Screen	User Feedback / Comments	Design Iteration (Changes Made)
Main Dashboard	"The sidebar is too plain and it's hard to find where to report a new problem quickly."	We added a high-contrast Orange "Report Issue" button in the top header and a navy blue sidebar with a user profile to make navigation clearer.
Report Issue Screen	"The dark screen makes it hard to see the input boxes, and I want to know if my photo is attached."	We changed to a clean white background and added a "Choose File" button that shows the file name once selected for confirmation.
Track Issue Screen	"I don't like seeing everyone else's reports in a big list; I only want to see my own progress."	We removed the public table and replaced it with a Reference Number Search Bar. Now, users only see the status of their specific report.
Admin Control Panel	"It's just a long list of text. As an admin, I need to see at a glance how many issues are still pending."	We added colored Statistics Cards (Blue, Orange, Green) at the top so the admin can instantly see the total, pending, and resolved counts.

1. MAIN DASHBOARD SCREEN

BEFORE



AFTER



Change Explained: Users missed the reporting function, so we made the button orange and moved it to the top right header for better visibility.

2. REPORT ISSUE SCREEN

BEFORE

CivicTrack (?) Help

< Jowe

Dashboard

Report Issue

My Reports

Updates

Report Issue

Issue Type: Water Leak (dropdown menu shows Pothole, Power Outage)

Description: Describe the issue in detail...

Location: Street address or landmark

Upload Photo

Submit

AFTER

CivicTrack - Report Issue

CivicTrack Alerts Help Dashboard

Dashboard

Report Issue

My Reports

Municipal Updates...

John Smith
Citizen

Log Out

Report a Municipal Issue

Issue Category: Pothole (dropdown menu)

Description:

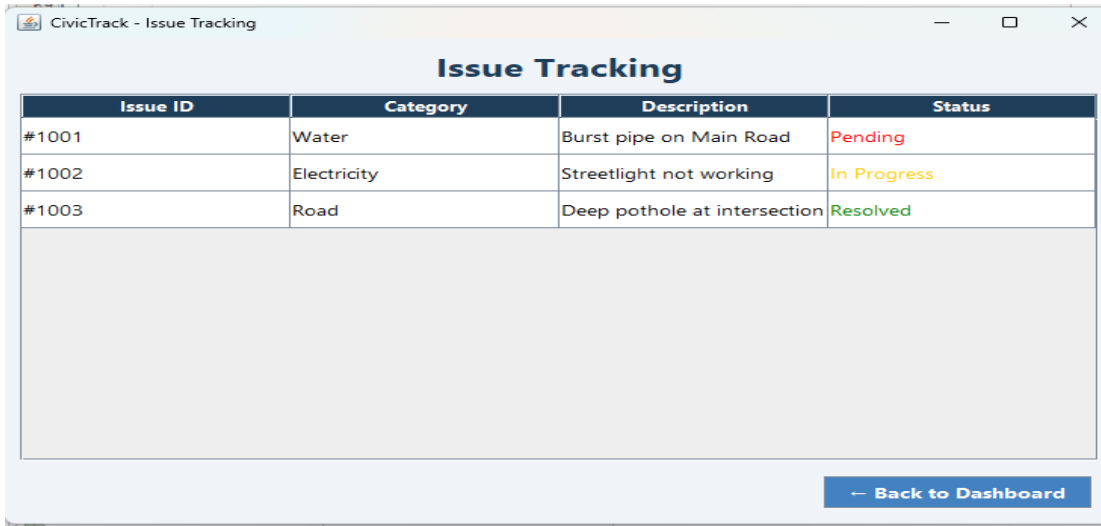
Location:

Upload Photo: Choose File No file selected

Change Explained: The original form was too cluttered, we used a white theme and added a photo upload button to provide evidence for technicians.

3. TRACK ISSUE SCREEN

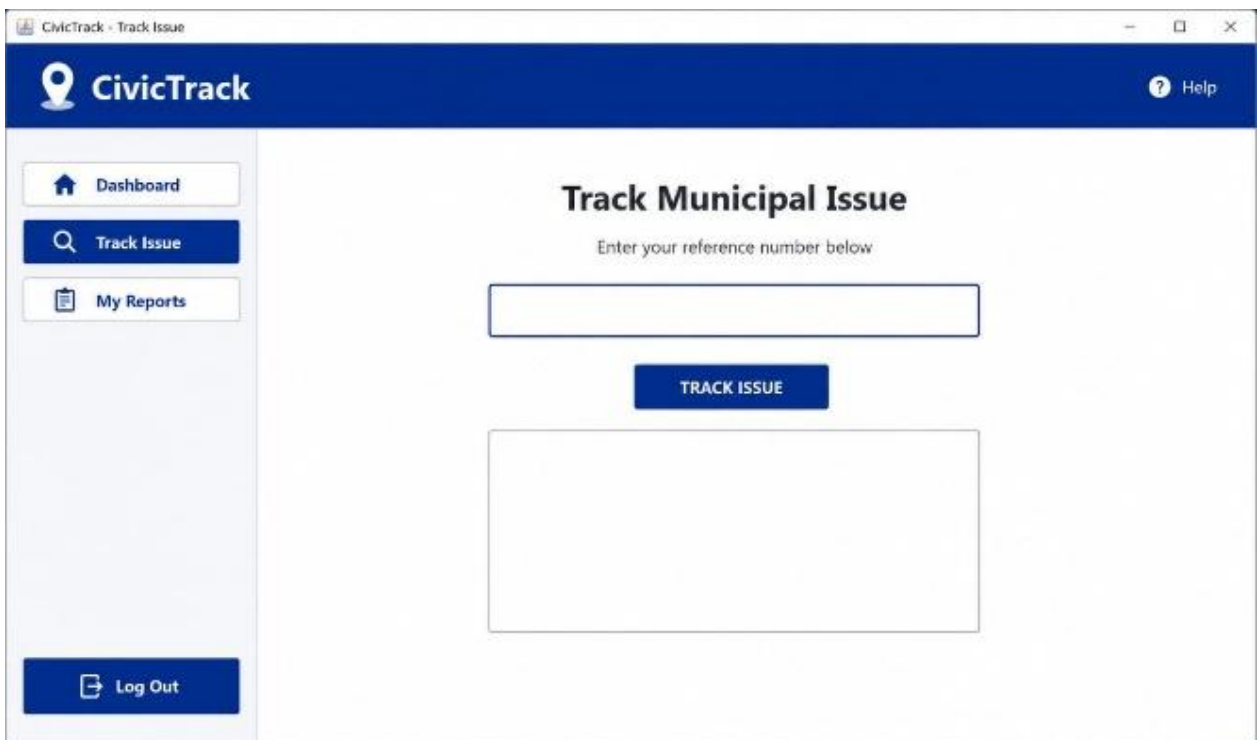
BEFORE



The screenshot shows a web application window titled "CivicTrack - Issue Tracking". The main content area is titled "Issue Tracking" and contains a table with four columns: "Issue ID", "Category", "Description", and "Status". The table lists three issues: #1001 (Water, Burst pipe on Main Road, Pending), #1002 (Electricity, Streetlight not working, In Progress), and #1003 (Road, Deep pothole at intersection, Resolved). Below the table is a large empty rectangular area. At the bottom right, there is a button labeled "Back to Dashboard".

Issue ID	Category	Description	Status
#1001	Water	Burst pipe on Main Road	Pending
#1002	Electricity	Streetlight not working	In Progress
#1003	Road	Deep pothole at intersection	Resolved

AFTER

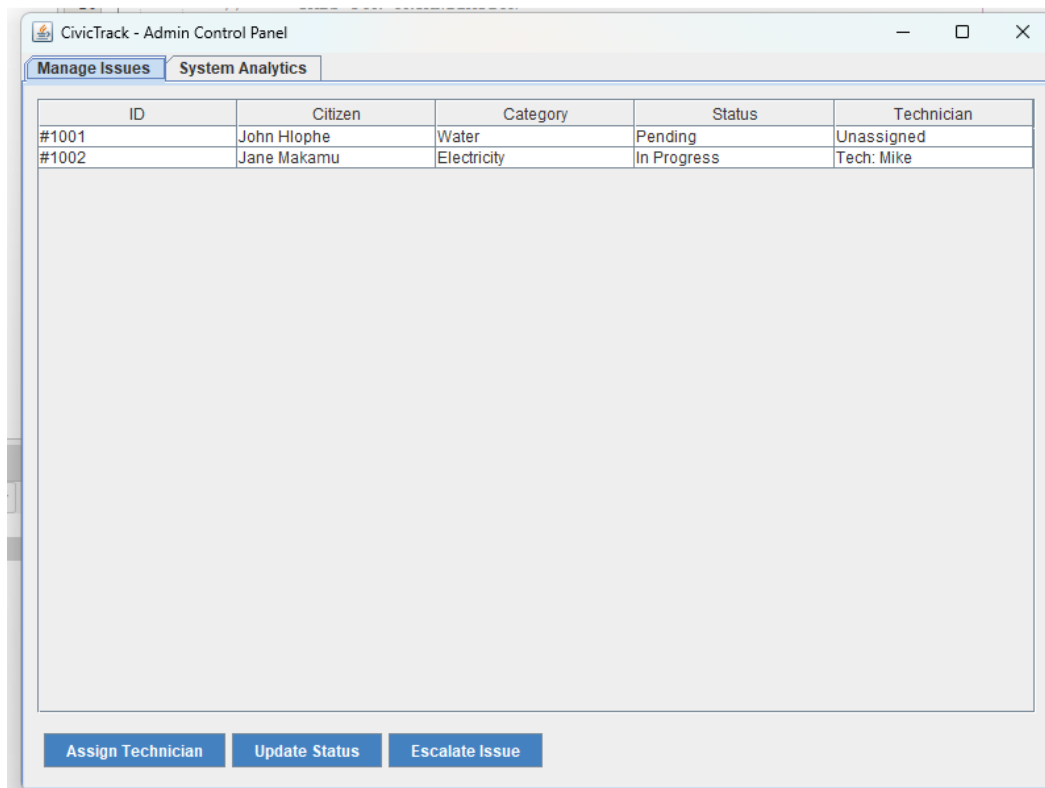


The screenshot shows a web application window titled "CivicTrack - Track Issue". The header bar is blue with the "CivicTrack" logo and a "Help" link. A sidebar on the left contains three buttons: "Dashboard", "Track Issue", and "My Reports". The main content area is titled "Track Municipal Issue" and contains a form. The form has a text input field with the placeholder text "Enter your reference number below". Below the input field is a blue button labeled "TRACK ISSUE". Below the button is a large empty rectangular area. At the bottom left of the sidebar, there is a "Log Out" button.

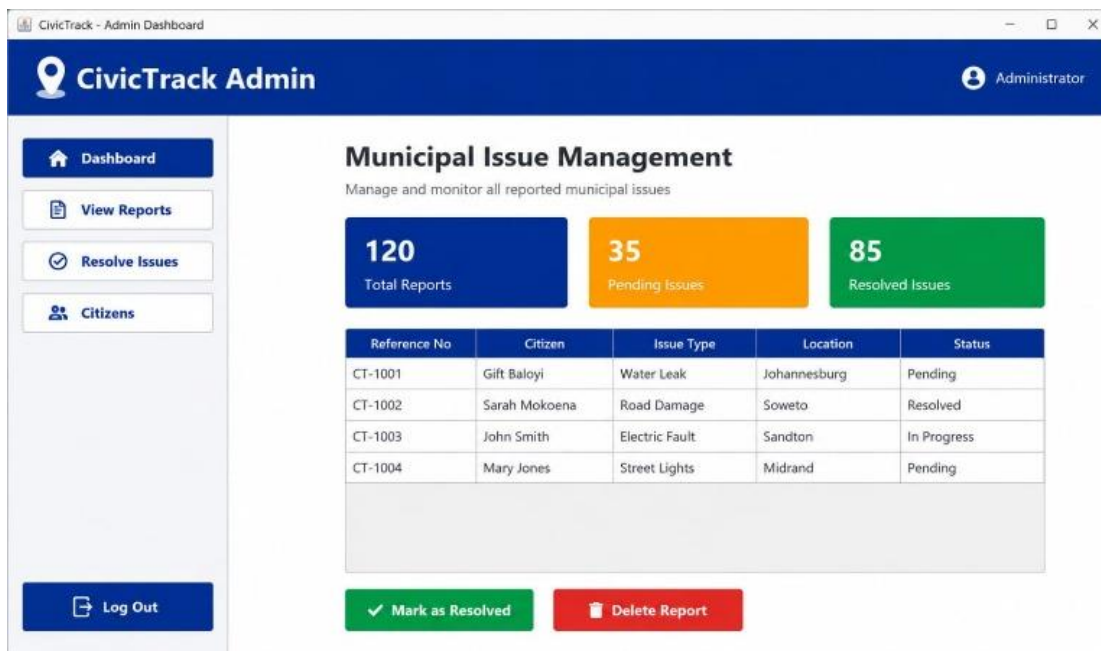
Change Explained: Testing showed users wanted more privacy, so we switched from a shared list to a private search field using a reference number.

4. ADMIN CONTROL PANEL SCREEN

BEFORE



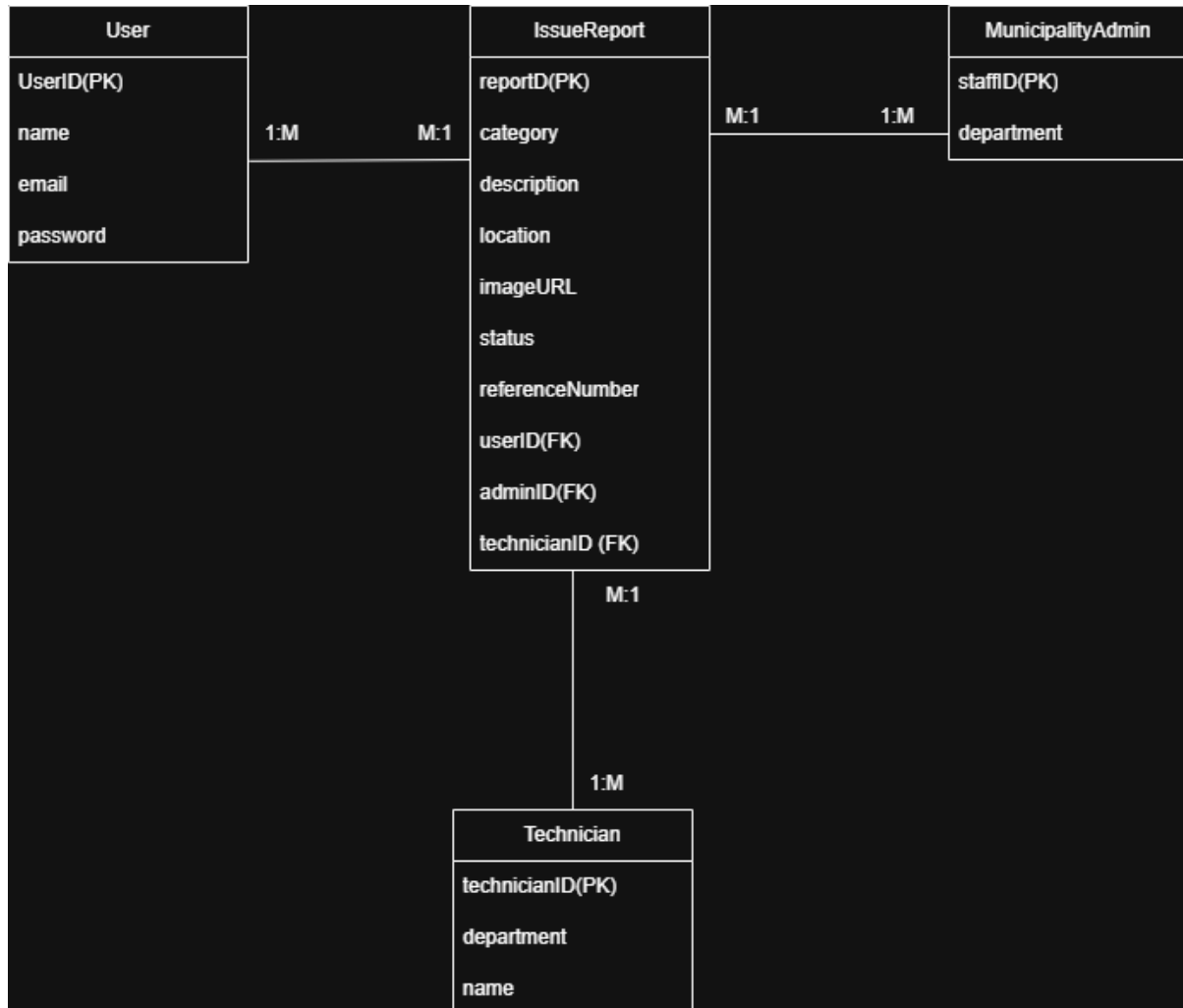
AFTER



Change Explained: Administrators found the list view slow to read, so we added summary cards to highlight urgent pending issues.

2. Database Design Components

2.1. Entity Relationship Diagram



2.2. Entity Descriptions

The following tables define the structure of the CivicTrack database, specifying data types and constraints for each attribute.

Table: User

Field Name	Data Type	Length	Constraints	Description
userID	INT	11	PK, Auto Increment	Unique identifier for each person.
name	VARCHAR	100	NOT NULL	Full legal name of the user.
email	VARCHAR	50	UNIQUE, NOT NULL	Unique login credential and notification address.
password	VARCHAR	255	NOT NULL	Hashed security credential.
role	VARCHAR	20	NOT NULL	Access level: Citizen, Admin, or Staff.

Table: IssueReport

Field Name	Data Type	Length	Constraints	Description
reportID	INT	11	PK, Auto Increment	Unique tracking number for the issue.
category	VARCHAR	30	NOT NULL	Type of issue (e.g., Water Leak, Pothole).
description	TEXT	N/A	NOT NULL	Detailed explanation of the problem.
location	VARCHAR	255	NOT NULL	Street address or landmark.

imageURL	VARCHAR	255	NULL	File path to uploaded photo evidence.
status	VARCHAR	20	DEFAULT 'Pending'	State: Pending, In Progress, or Resolved.
refNumber	VARCHAR	20	UNIQUE, NOT NULL	Alphanumeric code for citizen tracking.
userID	INT	11	FK (User.userID)	Links report to the submitting User.
adminID	INT	11	FK (Admin.staffID)	Links to the overseeing Admin.
technicianID	INT	11	FK (Tech.techID)	Links to the assigned repair staff.

Table: MunicipalityAdmin

Field Name	Data Type	Length	Constraints	Description
staffID	INT	11	PK, Auto Increment	Unique ID for municipal management.
name	VARCHAR	100	NOT NULL	Full name of the administrator.
department	VARCHAR	50	NOT NULL	Sector overseen (e.g., Water, Roads).

Table: Technician

Field Name	Data Type	Length	Constraints	Description
technicianID	INT	11	PK, Auto Increment	Unique ID for field repair worker.
name	VARCHAR	100	NOT NULL	Full name of the technician.

expertise	VARCHAR	50	NOT NULL	Field of technical skill (e.g., Electrical).
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2.3. Relationship Justification

The CivicTrack system utilizes a relational database model to ensure data consistency, prevent redundancy, and maintain strict accountability across municipal workflows.

- **User to IssueReport (1:M / One-to-Many):** A single **Citizen** can submit multiple unique reports over time; however, each **IssueReport** is linked to exactly one **userID**. This one-to-many relationship is critical for audit trails, ensuring that every reported leak or pothole is tied to a specific accountable individual for follow-up communications.
- **MunicipalityAdmin to IssueReport (1:M / One-to-Many):** Each **Admin** oversees a specific department (e.g., Water or Roads) and manages many different reports within that sector. By linking the **adminID** as a foreign key in the **IssueReport** table, the system ensures that every case has a dedicated supervisor responsible for status updates and escalation.
- **Technician to IssueReport (1:M / One-to-Many):** One **Technician** may be assigned a queue of multiple repair tasks based on their expertise, but each **IssueReport** is assigned to only one primary technician at a time. This prevents "collision" where two workers might attempt the same repair, ensuring clear responsibility and efficient resource allocation.
- **IssueReport to Image/Evidence (1:1 / One-to-One):** While your current schema stores the imageURL directly in the report table, this represents a **one-to-one** relationship logic. Each specific report is associated with one unique set of photo evidence that serves as a visual "receipt" of the problem, ensuring that the technician knows exactly what to fix before arriving on-site.

2.4. Technology Choice Justification

For the **CivicTrack** system, **MySQL** has been selected as the primary Relational Database Management System (RDBMS) to ensure the platform meets the rigorous data handling requirements of a municipal-scale application.

- **Prescribed Stack Alignment:** MySQL is the industry standard for integration with **Java** (the prescribed language for Phase 4). Using the **JDBC (Java**

Database Connectivity) driver ensures a stable and well-documented connection between our GUI and the data layer.

- **Support for ACID Properties:** To meet the **Reliability** requirements defined in Phase 2, MySQL provides ACID (Atomicity, Consistency, Isolation, Durability) compliance. This ensures that if a citizen submits a water leak report, the transaction is either completed fully or not at all, preventing "ghost" reports or corrupted data.
- **Relational Integrity:** The system relies heavily on **Foreign Key constraints** to link Users, Reports, and Technicians. MySQL's InnoDB engine provides the strict referential integrity needed to ensure a technician cannot be assigned to a report that does not exist.
- **Scalability for Municipal Data:** CivicTrack is designed for the city of Kimberley. MySQL efficiently handles the indexing required for high volumes of data, such as searching for all "Pending" issues across multiple wards without significant performance lag.
- **Cost-Effectiveness & Maintainability:** Being open-source, MySQL allows for a high degree of **maintainability**. It offers extensive community support and modular update capabilities, ensuring the system can be easily extended with new features in the future without expensive licensing hurdles.

2.5. Navigation Flow Summary

To ensure ease of use, the system follows a logical path:

- **Onboarding:** Users move from **Registration** to **Login**.
- **Core Action:** From the **Dashboard**, citizens can navigate to **Report Issue** or **Track Issue**.
- **Management:** Staff move from the **Dashboard** to the **Admin Control Panel** to perform updates.

References

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<https://app.diagrams.net/> (Accessed: 20 April 2026).

Kisten, M. (2026) 'NPRT63 Project (ICT 3rd Year): Lecture 2' [Lecture to ICT 3rd Year Students], *NPRT63*, Sol Plaatje University, 3 March.